

## Quiz 6: Metals

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Review of Session 6. ~15 minutes. Calculator needed.

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### Section A: Identifying the six metals

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1. Which metal on the list is reddish-orange?
2. Which metal on the list is magnetic?
3. Which two metals are much lighter (lower density) than the rest?
4. Which metal does **not** react with hydrochloric acid at all?
5. Give the atomic number: Mg, Al, Fe, Cu, Zn.

### Section B: Density

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1. Density = \_\_\_\_\_ ÷ \_\_\_\_\_.
2. What is the density of water in g/mL?
3. Convert 2.70 g/cm<sup>3</sup> to kg/m<sup>3</sup>.
4. A metal cube 2.0 cm on a side has a mass of 21.6 g. What is its density? Which metal is it?
5. A chunk of metal raises water from 15.0 mL to 24.0 mL and weighs 70.7 g. Density? Metal?

### Section C: Reactions, oxides, alloys

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1. Name the four basic reaction types.
  2. A metal dissolving in acid (  $\text{Zn} + 2 \text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$  ) is which type?
  3. Why does aluminum resist acid at first?
  4.  $2 \text{Mg} + \text{O}_2 \rightarrow 2 \text{MgO}$  is which reaction type?
  5. Name the metals in: steel, brass, bronze.
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